Infinite Sequences: An *infinite sequence* is an ordered collection of numbers $\{a_1, a_2, a_3, \ldots\}$, often denoted $\{a_n\}$, where n is called the *index* of the sequence.

Example: Consider the following sequences:

$$\{(-1)^n n\}, \qquad \left\{\frac{1}{n}\right\}, \qquad \{(-1)^n\}.$$

Function method does not work here because you can't extend a sequence in that way

- We say $\{a_n\}$ *converges* if $\lim_{n\to\infty} a_n = L$ (i.e. the limit exists).
- We say $\{a_n\}$ *diverges* if $\lim_{n\to\infty} a_n = \pm\infty$ or if the sequence oscillates forever (i.e. the limit does not exist).

Divergence: ∞, $-\infty$, oscillates

Function Method: If you want to determine the limit behavior of $\{a_n\}$, you can alternatively consider a function $f(x)$, $x \geq 1$ for which $f(n) = a_n$ for all $n = 1, 2, \ldots$. Then

- If $\lim_{x\to\infty} f(x) = L$, then $\lim_{n\to\infty} a_n = L$.
- If $\lim_{x\to\infty} f(x) = \pm\infty$, then $\lim_{n\to\infty} a_n = \pm\infty$.

Example: Determine whether the following sequences diverge or converge.

$$\left\{\left(1+\frac{2}{n}\right)^n\right\}, \qquad \left\{\frac{5n^5}{e^n}\right\}, \qquad \left\{\left(\frac{1}{3}\right)^n\right\}, \qquad \left\{\frac{n^2}{n^2+1}\right\}, \qquad \{\arctan(\ln(n))\}.$$

Raising using ln is helpful when you have exponentials

Let $f(x) = (1+\frac{2}{x})^x$

$\lim_{x\to\infty}(1+\frac{2}{x})^x$

$\neq 1$ since there is a battle between the infins

$e^{\ln((1+\frac{2}{x})^x)} = e^{x\ln(1+\frac{2}{x})}$ ↑ via log rules or power rules?

$= e^{\lim_{x\to\infty} \frac{\ln(1+\frac{2}{x})}{\frac{1}{x}}}$ $\frac{0}{0}$ you can then use lhopitals rule

Raised you can't evaluate exponents

$f' = e^{\lim_{x\to\infty} \frac{\frac{1}{1+\frac{2}{x}} \cdot \frac{-2}{x^2}}{\frac{-1}{x^2}}}$ why can you do this?

$= e^{\lim_{x\to\infty}\frac{2}{1+\frac{2}{x}}}$

$= e^2$, converges

Let $f(x) = (\frac{1}{3})^x$

$\lim_{x\to\infty}(\frac{1}{3})^x$

$e^{\ln((\frac{1}{3})^x)} = e^{x\ln(\frac{1}{3})}$

$e^{\lim_{x\to\infty} x\ln(\frac{1}{3})} = e^{\lim_{x\to\infty} x(-\ln(3))}$ $\ln(\frac{1}{3}) = -\ln(3)$ why?

$= e^{-\infty} = 0$ convergent

Let $f(x) = \frac{x^2}{x^2+1}$

$\lim_{x\to\infty}\frac{x^2}{x^2+1} \to \frac{\infty}{\infty}$, Lhopital

$\frac{2x}{2x} = 1$ 1, converges

Let $f(x) = \arctan(\ln(x))$

$\arctan(\lim_{x\to\infty}\ln(x))$

$\arctan(\ln(\infty)) \approx \arctan(\infty)$

What happens to arctan when it goes to $+\infty$?

$\frac{\pi}{2}$, converges - why?

Sequences / infinite sequences

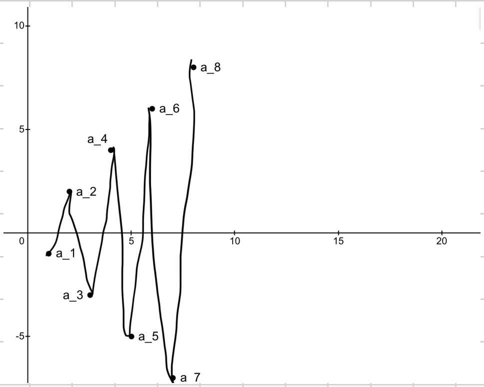
Ordered collection of real numbers $\{a_1, a_2, \ldots a_n\}$

Often denoted $\{a_n\}$ where n is the index

Ex $\{(-1)^n n\}$, $a_n = (-1)^n n$ $\qquad$ $\underset{a_1}{-1}, \underset{a_2}{2}, \underset{a_3}{-3}, \underset{a_4}{4}$ ← Divergent, $\lim_{n\to\infty}|a| = \infty$ but sign is also switching

$\{\frac{1}{n}\}$, $a_n = \frac{1}{n}$, $\qquad$ $\underset{a_1}{1}, \underset{a_2}{\frac{1}{2}}, \underset{a_3}{\frac{1}{3}}, \underset{a_4}{\frac{1}{4}}$ ← converges, denominator gets bigger so it goes to 0 ($\frac{1}{\infty} = 0$)

$\{(-1)^n\}$ $a_n = (-1)^n$ $\qquad$ $\underset{a_1}{-1}, \underset{a_2}{1}, \underset{a_3}{-1}, \underset{a_4}{1}$ ← oscillates forever → divergent



What about $\{(-\frac{1}{3})^n\}$?

Will it oscillate? That doesn't matter because it still converges to 0 in this case

Absolute value theorem

If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$. This only works if the limit is 0.

Absolute Value Theorem: If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$. Note that this only works if the limit is zero.

Example: Determine whether the following sequences diverge or converge:

$$\left\{\left(-\frac{1}{3}\right)^n\right\}, \qquad \left\{(-1)^n \frac{n^2}{n^2+1}\right\}.$$

0, see prev. pages

$\lim_{n\to\infty} \left|(-1)^n \frac{n^2}{n^2+1}\right| = 1 \neq 0$

- If all n even, $\lim_{n\to\infty} \underbrace{(-1)^n}_{\text{-1 to an even number is always 1}} \frac{n^2}{n^2+1} = 1$
- If all n odd, $\lim_{n\to\infty} \underbrace{(-1)^n}_{\text{-1 to an odd number is always odd}} \frac{n^2}{n^2+1} = -1$

This is kind of like approaching a limit from both sides $1 \neq -1$ so since from both sides it's not the same, the limit DNE

Helpful Limit: For the sequence $\{r^n\}$ where r is a real number,

$$\lim_{n\to\infty} r^n = \begin{cases} 0, & \text{if } |r| < 1 \\ \infty, & \text{if } |r| > 1 \end{cases}$$

$|r|$ is less than 1, $\lim_{n\to\infty} r^n = 0$

$|r|$ is greater than 1, $\lim_{n\to\infty} r^n = \pm\infty$

$\lim_{n\to\infty} r^n = 1$ if $r = 1$

Example: Determine whether the following sequences,

$$\left\{\left(\frac{\sqrt{2}}{2}\right)^n\right\}, \qquad \{3^n 8^{-n}\}, \qquad \{4^{n-2} 8^{-n}\}.$$

$a^n = 3^n 8^{-n} = \left(\frac{3}{8}\right)^n$

$a = \frac{3}{8} \to \frac{3}{8} < 1 \to 0$

$\to a_n = 4^n 4^{-2} 8^{-n} = \frac{1}{16}\left(\frac{4}{8}\right)^n$

what about this part?

$\frac{4}{8} < 1, \to 0$

converge or diverge.

Squeeze Theorem: If $a_n \le b_n \le c_n$ for all $n = 1, 2, \ldots$ and

$$\lim_{n\to\infty} a_n = L = \lim_{n\to\infty} c_n,$$

then $\lim_{n\to\infty} b_n = L$ also.

Example: Determine whether the following sequence,

$$\left\{(-1)^n \frac{\cos(n)}{n^2}\right\}$$

converges or diverges.

$a_n = \left(-\frac{1}{2}\right)^n \cos(n)$

$|a_n| = \frac{|\cos(n)|}{2^n}$

$\lim_{x\to\infty} \frac{\cos(x)}{2^x}$ can't say anything about this

$\lim_{n\to\infty} \frac{|\cos(n)|}{2^n}$ We can't say much about this yet. We can use the Squeeze theorem, as the function method does not help here

$\lim_{n\to\infty} \left(\frac{1}{2}\right)^n = \lim_{n\to\infty} -\left(\frac{1}{2}\right)^n = 0$

Is this because of the absval thing?

$-\left(\frac{1}{2}\right)^n \le b_n = \left(-\frac{1}{2}\right)^n \cos(n) \le \left(\frac{1}{2}\right)^n = c_n$

So $\lim_{n\to\infty} \left(-\frac{1}{2}\right)^n \cos(n) = 0$

Montone Sequences: We say a sequence is *monotone* if it does not oscillate. More specifically, a sequence $\{a_n\}$ is:

- *Monotonically Increasing* if $a_n \le a_{n+1}$ for all $n = 1, 2, \ldots$
- *Monotonically Decreasing* if $a_n \ge a_{n+1}$ for all $n = 1, 2, \ldots$

Bounded Sequences: We say a sequence $\{a_n\}$ is:

- *Bounded Above* if there is a number M such that $a_n \le M$ for all $n = 1, 2, \ldots$
- *Bounded Below* if there is a number m such that $a_n \ge m$ for all $n = 1, 2, \ldots$

If both statements above are true, we say $\{a_n\}$ is a *bounded sequence*.

Example: Consider

$$\left\{\frac{n}{n+1}\right\}.$$

Does this sequence converge?

Ignore for now

Theorem: Every bounded monotone sequence converges.